



Advanced language features

- Macros and tricks for your instrument...

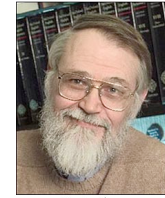


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DECLARE / INITIALIZE / USERVARS %{ } sections



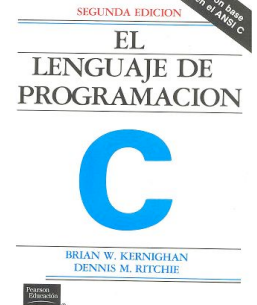
K & R. / GNU



Brian Kernighan



Dennis Ritchie



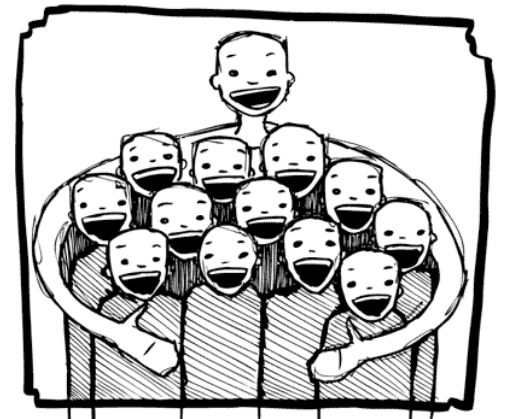
- Use the DECLARE section define user variables and functions.
 - **DECLARE** %{
 - double some_global_var;
 - %}
- Use INITIALIZE for initialization of user variables and calculations.
 - **INITIALIZE** %{
 - myvar = sqrt(PI*input_var)*rand01();
 - %}
- - Both use normal c-syntax.
- Use USERVARS for particle-dependent flags (**McStas >=3 specific**)
 - **USERVARS** %{
 - double myvar;
 - %}
- - Declarations of style “type varname;” only!
- BEWARE: What you do in the c-style areas is c-standard, e.g. trigonometric functions from math.h use radians! - McStas ROTATED specifiers work in degrees, etc...





%include

- Instrumentfiles can include external c-code or other instrumentfiles... See the examples:
- ILL_H15_IN6.instr:%include "monitor_nd-lib"
- ILL_H16_IN5.instr:%include "ILL_H16.instr"
- ILL_H25_IN22.instr:%include "ILL_H25.instr"
- ILL_H25_IN22.instr:%include "templateTAS.instr"
- May be used in the DECLARE / INITIALIZE / TRACE / FINALLY / SAVE sections



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Instruments may depend on external commands / linker instructions / searches etc.

- SHELL “some command run prior to code generation “
 - (No good example use at the moment)
- SEARCH “/add/a/component/search/directory/“
 - Used mainly in Greg Tucker’s mcstas-antlr infrastructure for BIFROST that uses special repos
- SEARCH SHELL “wget https://some/component/file.comp“
 - Used mainly in Greg Tucker’s mcstas-antlr infrastructure for BIFROST that uses special repos
- DEPENDENCY “-DSOME_DEFINE -lsome -L/some/lib -I/some/include “
 - Used e.g. for linking with MCPL and NCrystal libs, NeXus libs etc.
 - Raised both by comp and instrument level code

COMPONENT instance syntax in one, complex view...



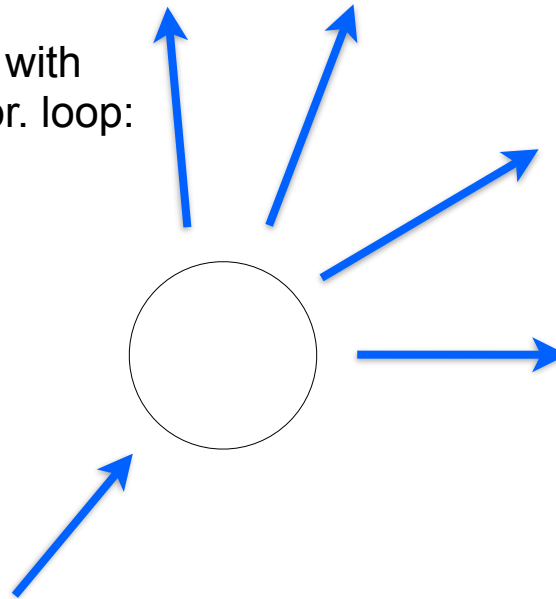
```
{SPLIT} COMPONENT name = comp(parameters) {WHEN condition}
AT (...) [RELATIVE [reference|PREVIOUS] | ABSOLUTE]
{ROTATED {RELATIVE [reference|PREVIOUS] | ABSOLUTE} }
{GROUP group_name}
{EXTEND C_code}
{JUMP [reference|PREVIOUS|MYSELF|NEXT] [ITERATE number_of_times | WHEN condition] }
```





SPLIT

- Increase statistics beyond this point in the instrumentfile
- SPLIT n MyArm = Arm()
- AT somewhere
- will “formulate an for-statement” with a copy of the incoming particle pr. loop:
 - for j=1:n
 - comp1
 - comp2
 - comp3
 - ...
 - end (of instrument)
- ONLY meaningful in case of Monte Carlo choices after SPLIT point...



Problem: McStas Single_crystal.comp “slow” for large unit cell diffraction studies

- Example: Rubredoxin 1 timebin, 1000 x,y-bins

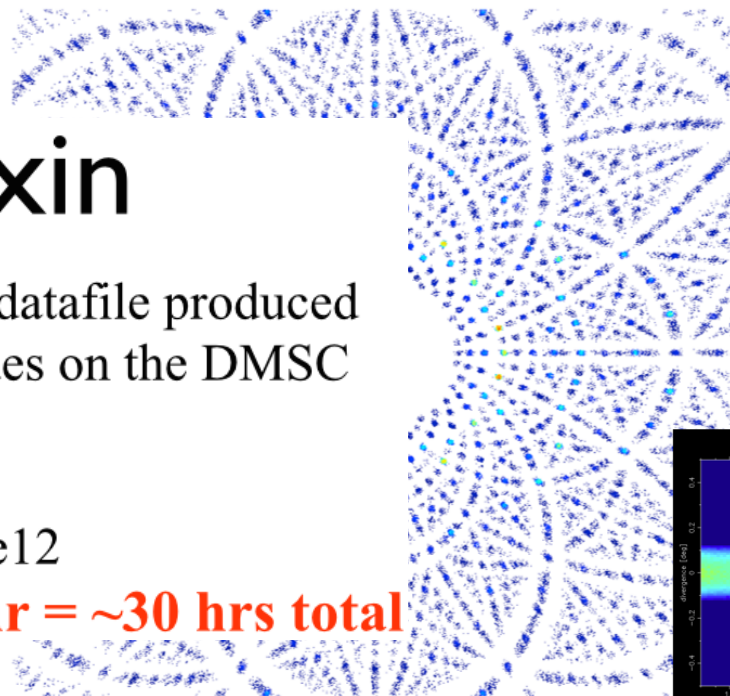
Rubredoxin

Images created from simulated datafile produced
August 20th 2012 using 25 nodes on the DMSC
cluster.

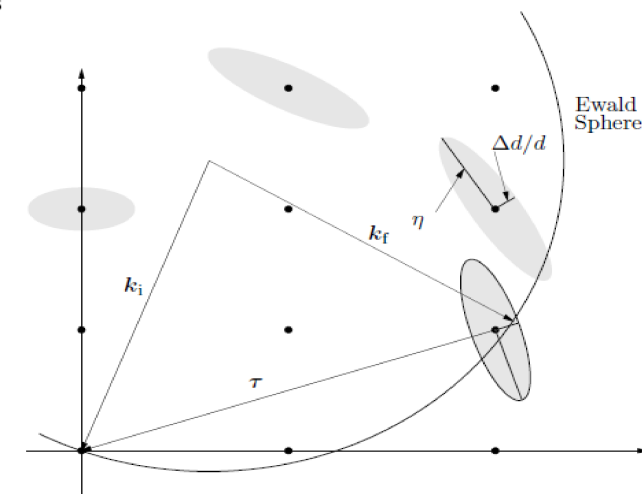
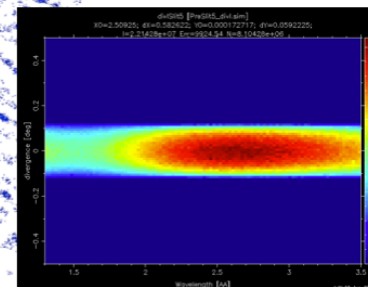
Neutron count: 1e12

Simulation time: ~10 + ~20 hr = ~30 hrs total

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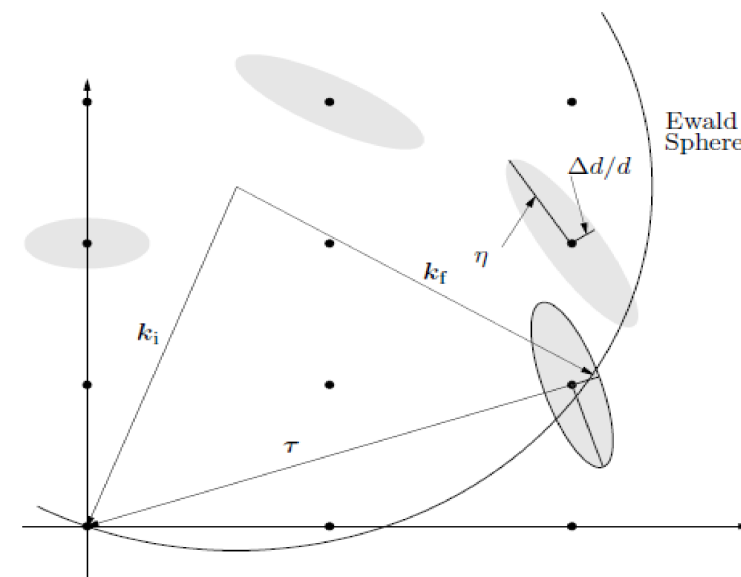
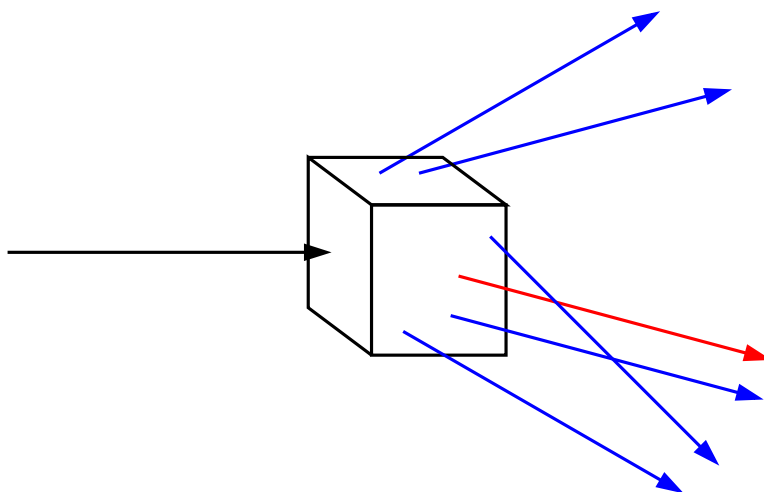


Neutroncount: 1e12
No gravitation
Xtal size: 0.5 mm
Xtal mosaicity: 12'
Detector: 50 x 50 cm flat
Detector-to-sample
distance: 20 cm
Guide length: 131 m
Guide dimensions: 9.5 cm
 $\lambda_{\min} = 1.3 \text{ \AA}$
 $\lambda_{\max} = 3.5 \text{ \AA}$
Timespan: 51.39 to 143.4 ms
Divergence = 0.2 degs



Algorithm improvement: Use incoming neutrons more efficiently - scatter each one on all possible reflections

- **Red**: Original algorithm, one incoming neutron used only once
- **Blue**: Improved algorithm, each incoming neutron scattered (via SPLIT keyword) all possible times
- Component makes **estimate on average number of “active” diffraction spots** - in the case Rubredoxin this is around **50!**



!!! For now, does not work in our GPU implementation !!!



GROUP - components working in parallel



*AT (0,0,-LMM) RELATIVE Cradle ROTATED (0,A1/2,0) RELATIVE Cradle
GROUP IN6Monoks*

*AT (0,0,0) RELATIVE Cradle ROTATED (0,A2/2,0) RELATIVE Cradle
GROUP IN6Monoks*

*- One comp after the other is “tried” in sequential order until the neutron was
SCATTERED.*

This is an XOR

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EXTEND

- Enrich component behaviour using EXTEND:

```
COMPONENT Mono1 = Monochromator_curved(...)
AT (0,0, -LMM) RELATIVE Cradle ROTATED (0,A1/2,0) RELATIVE Cradle
GROUP IN6Monoks
```

```
EXTEND
```

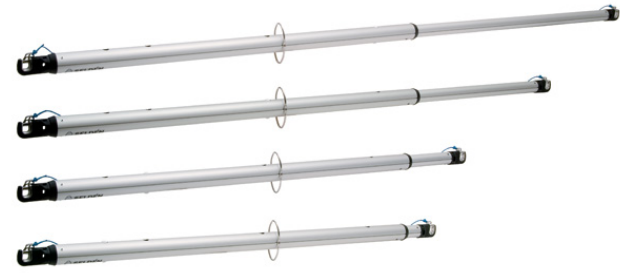
```
%{
  if (SCATTERED) { myvar = 1; }
}%
```

```
...
```

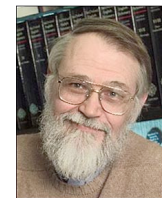
```
COMPONENT Mono2 = Monochromator_curved(...)
AT (0,0, 0) RELATIVE Cradle ROTATED (0,A2/2,0) RELATIVE Cradle
GROUP IN6Monoks
```

```
%{
  if (SCATTERED) { myvar = 2 ;}
}%
```

Injected “per instance” post component TRACE



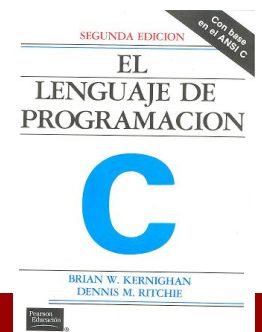
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WHEN

- Syntax:

COMPONENT Mine = Yours(blah, blah)

WHEN (c-expression) AT (....)

- Is very powerful when combined with EXTEND and user variables, or as a method to let input parameters select if certain components are active.
- Example: Use EXTEND to flag if neutron was scattered on one monochromator blade or another. Then later use WHEN to only show contribution from blade N at sample position?

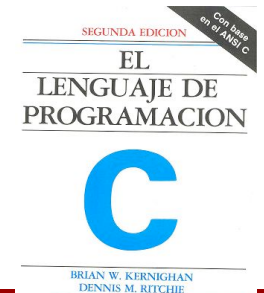
COMPONENT Mon = PSD_monitor(...)

WHEN (myvar==1) AT (0,0,0) RELATIVE Sample

Conditional component TRACE, takes any c-expr



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JUMP

- A goto. Be careful. Can be used in two situations:
 - JUMP to myself
 - JUMP to an Arm
-
- No coordinate transformations are applied... (Meaning that if the Arms you JUMP between do not coincide you will “move” / “reorient” the neutrons...)
-
- Syntaxes:
 - COMPONENT a=b(...)
 - WHEN (expr) AT (...) JUMP somewhere
-
- COMPONENT a=b(...)
 - WHEN (expr) AT (...) JUMP myself



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BEWARE - This IS a



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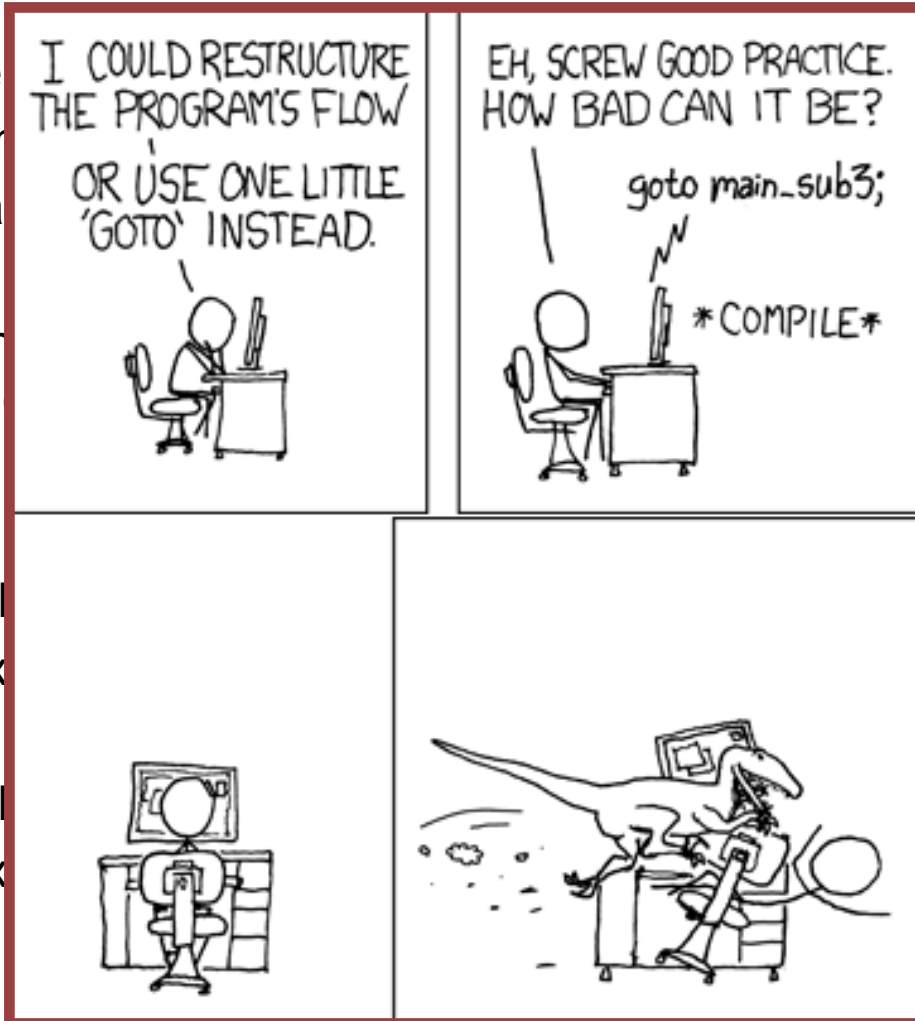
JUMP

- A goto. Be
- JUMP to m
- JUMP to a

- No coordin
- between d

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- COMPONENT
- WHEN (ex

- COMPONENT
- WHEN (ex



BEWARE - This IS a

that if the Arms you JUMP
e neutrons...)



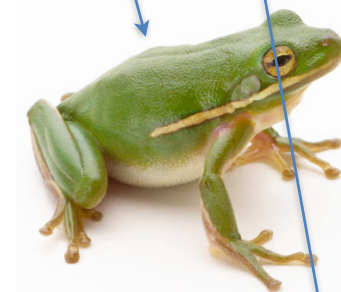
Just don't? :-)

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COPY- inside instruments

- In instruments: (see ILL_H25.instr)
- COMPONENT H25_1 = Guide_gravity(
 - $w1=0.03$, $h1=0.2$, $w2=0.03$, $h2=0.2$, $l=L_H25_1$,
 - $R0=gR0$, $Qc=gQc$, $\alpha=g\alpha$, $m=m$, $W=gW$)
 - AT (0,0, $A_Thickness+gGap$) RELATIVE PREVIOUS
 - ROTATED (0, $Rh_H25_1,0$) RELATIVE PREVIOUS
- COMPONENT COPY(H25_1) = COPY(H25_1)
 - AT (0,0, L_H25_1+gGap) RELATIVE PREVIOUS
 - ROTATED (0, $Rh_H25_1,0$) RELATIVE PREVIOUS
- COMPONENT COPY(H25_1) = COPY(H25_1)($W=2*gW$)
 - AT (0,0, L_H25_1+gGap) RELATIVE PREVIOUS
 - ROTATED (0, $Rh_H25_1,0$) RELATIVE PREVIOUS



Example: Decompose multiple scattering from Single_crystal



```

USERVARS %{
  double multiple_scatt;
%}

...
COMPONENT Crystal = Single_crystal(... order=0 ...)
AT (0,0,0) RELATIVE somewhere
EXTEND %{
  multiple_scatt=SCATTERED;
%}

...
COMPONENT PSD_single=PSD_monitor(...)
WHEN (multiple_scatt==1) AT (0,0,0) RELATIVE somewhere_else

COMPONENT PSD_multiple=PSD_monitor(...)
WHEN (multiple_scatt > 1) AT (0,0,0) RELATIVE somewhere_else

```

**Or use with
Monitor_nD
user1="multiple_scatt"**

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